Two oriented square-torus amalgamations

Human Compression project

The fixed Schulte–Weiss survey attributes the following two examples to Hubard–Weiss, *Self-duality of chiral polytopes* (2005), p. 132. We verify their presented groups and the actual intersection and chirality conditions. This is a recovery of two established examples, not a new existence result or a classification of toroidal amalgamations.

Write $T(s,t) = \{4,4\}_{(s,t)}$ with its specified orientation, and put

$$R(x,y;s,t) = (x^{-1}y)^s(xy^{-1})^t.$$

We retain two precise standard interfaces stated in the survey: the oriented square-torus presentation on rotations x, y is $x^4 = y^4 = (xy)^2 = R(x,y;s,t) = 1$; and the rank-four rotation reconstruction criterion consists of the string rotation relations and the three intersection equalities below. The square-torus convention is the lattice spanned by (s,t) and $(-t,s)$. In the cases here its index is ten, so its rotation group has order forty. No completeness assertion for other parameters is used.

Proposition 1. *For $(s,t) = (1,3)$ or $(3,1)$, let*

$$G(s,t) = \langle u,v,w \mid u^4 = v^4 = w^4 = (uv)^2 = (vw)^2 = (uvw)^2 = 1,$$
$$R(u,v;s,t) = R(v,w;1,3) = 1\rangle.$$

These are the automorphism groups of the oriented universal chiral amalgamations with facets $T(s,t)$ and vertex figures $T(1,3)$. Their orders are respectively 2000 *and* 960*; hence the two resulting polytopes are not isomorphic.*

Proof. The supplied finite calculation constructs an isomorphism from each full finitely presented group to a permutation group, of degrees fifty and twenty respectively. This is not merely a finite quotient representation. The following quantities are checked in those actual permutation groups:

(s,t)	$\lvert G(s,t)\rvert$	$\lvert\langle u,v\rangle\rvert$	$\lvert\langle v,w\rangle\rvert$	$\lvert\langle u,v\rangle \cap \langle v,w\rangle\rvert$
$(1,3)$	2000	40	40	4
$(3,1)$	960	40	40	4

Each of u, v, w has order four, and

$$\langle u\rangle \cap \langle v\rangle = \langle v\rangle \cap \langle w\rangle = 1, \qquad \langle u,v\rangle \cap \langle v,w\rangle = \langle v\rangle.$$

The local finitely presented groups are separately checked to have order forty. Their natural epimorphisms onto the actual end subgroups are therefore isomorphisms, so the prescribed toroids do not collapse. The retained reconstruction criterion now supplies a rank-four polytope with precisely those oriented local types.

The regularity test would require an automorphism taking

$$(u,v,w) \longmapsto (u^{-1}, u^2v, w).$$

After this substitution, both displayed toroidal relators are nonidentity in each group. Such an automorphism cannot exist. The reconstructed polytopes are therefore chiral and $G(s,t)$ is their full automorphism group, rather than only the rotation subgroup of a regular polytope.

The two added relators are exactly the local defining relations beyond the standard rotation relations. Every oriented amalgamation with these specified facets and vertex figures consequently has its marked rotation group as a quotient of $G(s,t)$. Together with the verified existence and uncollapsed local types, the retained oriented amalgamation interface gives the stated universal object. Finally unequal automorphism-group orders exclude an isomorphism between the two polytopes. □

Verification and attribution. The executable calculation uses GAP's isomorphism algorithm for finite presentations. It checks bijectivity and all defining relations, and exports the three literal permutations. A separate Python calculation, using only composition and exhaustive closure of the resulting small groups, verifies both group orders, every stated subgroup intersection, local orders and both failed reflection relators. This second calculation does not itself prove the upper order of the finite presentation; that step remains the explicit GAP computation. A damaged marking is rejected. These are computational checks, not Lean certificates.

The source is the fixed arXiv edition math/0608397v1, Section 4 (physical p. 7) for the orders and attribution, and Section 5 (physical p. 12) for the literal toroidal relator. The rank-four reconstruction and reflection criteria occur in Section 2 (physical p. 4), attributed there to Schulte–Weiss, *Chirality and projective linear groups* (1994). The cited 1994 and 2005 primary papers were not independently inspected for this component; the exact interfaces above are retained from the read survey, while the finite group and intersection calculations are reproduced here. General chiral amalgamation existence or finiteness questions are not resolved by these two examples.